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Weather-resilient Object Detection Framework for Autonomous Vehicles Using Conditional Preprocessing and YOLOv8

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Received: 19 June 2025 / Revised: 6 April 2026 / Accepted: 27 May 2026
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Abstract

Autonomous vehicles encounter serious difficulties in adverse weather conditions such as fog, snow, rain, and sandstorms, where image quality is reduced and object detection becomes less reliable. Many existing approaches perform poorly in such environments because they are trained mainly on clear-weather datasets and rely on generic preprocessing methods. This study presents a framework that combines weather-based preprocessing, image enhancement, and optimized object detection. Weather conditions are first identified using ResNet-18. Based on the detected condition, the right preprocessing methods, like dehazing and denoising, are used to make the image clearer. ESRGAN is then used to improve the processed images by bringing back important details that are needed for detection. Bayesian optimization is used to make the YOLOv8 model better at finding objects in real time. The results show improved performance across different weather conditions. The model achieves precision values of 81.08% in snow, 85.64% in rain, 85.44% in fog, and 65.76% in sand. The corresponding mAP values are 75.95% for snow, 70.80% for rain, 75.35% for fog, and 49.50% for sand. These results show that the system can still find things even when conditions change. The framework makes it easier to find objects in difficult places and makes it safer for self-driving cars to work.

Keywords Autonomous vehicles · Object detection · YOLOv8 · ESRGAN · Weather classification · Bayesian optimization

Abbreviations

DAWN	Detection in adverse weather nature
ESRGAN	Enhanced Super Resolution Generative Adversarial Network
RRDB	Residual-in-Residual Dense Blocks
YOLOv8	You only look once version 8
ResNet	Residual Neural Network
LIDAR	Light detection and ranging
ReLu	Rectified linear unit
PSNR	Peak signal to-noise ratio
SSIM	Structural similarity index measure



Published online: 22 June 2026

SGD	Stochastic gradient descent
DFL	Distribution focal loss
DCP	Dark Channel Prior
RNN	Recurrent Neural Networ
CLAHE	Contrast-limited adaptive histogram equalization

1 Introduction

Experts say that self-driving cars will make transportation safer, better at managing traffic, and more energy-efficient. However, they can't be used in bad weather like fog, rain, snow, and sandstorms because it's hard to drive them in those conditions. These conditions make visual data less clear and less accurate. Fog makes things harder to see by scattering light, and snow and sandstorms add noise and distortion that make it hard to tell what things are [1]. These effects impact safety because autonomous systems depend on accurate perception for navigation and real-time decisions [2]. Addressing these issues is necessary for reliable operation in real-world environments.

This study introduces a framework that integrates weather classification, condition-specific preprocessing, image enhancement, and optimized object detection. A ResNet-18 model figures out what the weather is like and tells you what to do to improve the quality of the image, such as dehazing, denoising, and reducing snow [3]. ESRGAN improves the resolution of images and keeps important details [4]. YOLOv8 can find objects in real time, and Bayesian optimization makes it more accurate and reliable in different weather conditions [5]. This method lets you adapt to different environments and makes self-driving cars more useful.

This paper is organized as follows. The abstract provides an overview of the purpose and results. The introduction highlights the rationale for conducting the study and the process adopted. The literature review explores past approaches to solving the problem. The methodology section presents the proposed solution. The experimental results section evaluates the performance of the proposed model.

1.1 Challenges in Object Detection Under Adverse Weather

Detection tasks become difficult when there are poor weather conditions like fog, rain, snow, and sandstorms since the image quality gets affected, and thus the performance of the model will be poor. Low contrast is another issue because the objects and the background look alike, making it hard to see the edges of the objects or identify small objects or those far away. There is decreased visibility in cases where there is thick fog or heavy rain since the distance within which objects can be seen is reduced and they appear faint or partially visible. Rain streaks, snow particles, and flying sand due to the mentioned weather cause noise in the images and may affect the model's ability to make the correct predictions.

1.2 Major Contributions

The main contributions of this paper are listed below:

- A weather-aware object detection framework is introduced. The system first detects the weather condition and then applies suitable preprocessing before object detection.
- A conditional preprocessing method is designed to handle fog, rain, snow, and sandstorm images separately instead of using the same method for all images.
- ESRGAN is used to improve image clarity and restore important details before passing the image to the detection model.

- YOLOv8 is used as the object detection model, and its performance is improved through proper hyperparameter tuning.
- The proposed framework is tested on the DAWN dataset, and results are analyzed for each weather condition.
- An ablation study and failure case analysis are provided to show the strengths and limitations of the framework.

2 Related Work

The field of weather-resilient object detection for autonomous vehicles has made significant progress in addressing challenges posed by adverse weather. Recent research is categorized into object detection frameworks, weather-related challenges, dataset improvements, and optimization strategies.

2.1 Object Detection Frameworks

In recent years, developments in CNNs and transformer architectures have made improvements in object detection regarding accuracy and computational efficiency. Transformers that incorporate attention mechanisms along with CNNs have shown excellent results in dynamically changing environments like the BDD dataset, enabling real-time object detection [6]. DenseNet architectures are highly successful in traffic classification tasks; however, they require a high computational load, making them unsuitable for real-time applications [7]. YOLO architectures, especially YOLOv3, have been widely used for real-time object detection because of their speed, and changes in grid sizes can further improve small-object detection [8]. Techniques such as Non-Maximum Suppression (NMS) improve detection accuracy, especially in systems with limited resources [9].

Detection techniques like NMS contribute to higher precision in detection, especially in systems that operate under limited resource availability [9]. Current research efforts focus on enhancing detection capabilities in adverse environmental settings. In one study, Hasanujjaman et al. [10] developed an Adaptive Fusion Alignment (AFA) framework combined with a Deep Convolutional Neural Network (D-CNN) model, which increases detection accuracy while decreasing error rate in unfavorable circumstances. The above example illustrates how adaptive control and learning-based prediction can be applied to motion planning tasks. Another paper tackles the issue of generalization in novel environmental contexts [11], while another focuses on detection enhancement in dynamic scenarios through the use of AI-based classification techniques [12]. Hybrid approaches like SD-YOLO-AWDNet have been devised for the purpose of detecting objects in adverse weather situations [13]. Geometric constraints obtained through LIDAR 3D object detection are also used in scenarios where visual information is distorted [14].

2.2 Weather-related Challenges and Enhancement

Fog, rain, snow, and sandstorms affect images in different ways, and one preprocessing method cannot handle all conditions effectively [1, 2, 15].

Fog. Fog disperses light, leading to a decrease in contrast, causing edges to be blurred. In order to restore the visibility of objects, DCP makes use of the atmospheric lighting and transmission values to compensate for the hazy effect; unfortunately, it saturates the areas in the sky [3, 16]. On the other hand, PMDNet improves on this concept by retaining the spatial and context information at all the levels of processing [3]. After haze removal, distant objects become hard to see, resulting in poor recall [17]. LIDAR is also affected by backscatter, which lowers point cloud quality [1].

Rain. Rain leads to artifacts such as streaking, wet surface reflections, and droplet artifacts that impair the detection capability of small objects (pedestrians and two-wheelers) [1, 18]. Severe rain impacts the effectiveness of radars, reducing the effective detection range by up to 45% [1]. The RNN-based de-raining method can

distinguish rain artifacts from other features in the image using temporal features [19]. Detection using YOLOv9 and rain augmentation provides an improvement in real conditions [18]. The use of sensor fusion that incorporates both camera and radar sensors allows for reliable detection even if the sensors alone are not able to function [10, 15].

Snow. Snow affects the process of detection in two ways, first by interfering during fall and secondly when it has accumulated. The first case introduces interference similar to that of rain while in the latter case, it results in less texture and blending of objects into the background [1, 2]. There is an increase in false negative results for models trained in clear weather [20]. Methods such as CLAHE with bilateral filtering help restore edge details before detection [19]. LIDAR is also affected by snow due to false reflections from particles [14].

Sandstorm. Sandstorms lead to the formation of thick haze where there is a varying amount of intensity due to the particle movement. The conventional method of DCP fails since sand colored areas violate its principles [15]. The images often suffer from the yellowish hue, thus reducing the distinction between objects and the environment [2, 13]. There is an SD-YOLO-AWDNet approach that includes the usage of a particular weather module for sandstorms, but it performs worse compared to fog or rain, as a result of a lack of data [13]. LIDAR-based detection with geometric constraints helps when camera input becomes unreliable [14].

Enhancement Approaches. ESRGAN restores fine image details lost due to weather degradation and improves detection performance when combined with YOLO-based models [4, 19]. Applying preprocessing methods based on the detected weather condition improves results compared to a fixed pipeline [21]. Sensor fusion using camera, radar, and LIDAR improves reliability when individual sensors are affected [15, 22].

2.3 Dataset Improvements

Weather-specific datasets such as the Canadian Vehicle Dataset (CVD) improve detection performance in challenging environments [23]. These datasets perform better than general datasets such as RoboFlow or COCO for weather-based evaluation [17]. Data augmentation methods that simulate adverse weather conditions improve model robustness during training [18]. However, many datasets do not include extreme weather scenarios, which limits model generalization [20].

2.4 Optimization Strategies and Limitations

Optimization techniques such as Bayesian optimization improve model performance under all weather conditions [5]. The problem of balancing computational efficiency and accuracy in real-time applications persists. Although YOLOv3 operates in real time, it is less accurate in harsh weather conditions [24]. The accuracy-speed trade-off in real-time solutions is prevalent [25]. Optimization techniques and weather-specific training are not common practices [16]. Pipeline integration of ESRGAN-CNN-based denoising algorithms is rare, affecting their robustness [26]. These are important challenges that must be addressed.Addressing these issues is important for reliable real-time detection in different weather conditions [14].

However, hyperparameters' optimization procedures, for instance, Bayesian optimization, improve the model's behavior in different weather conditions [5]. Balancing the speed of the system at the moment and its accuracy remains a difficult task. Models like YOLOv3 have good speed characteristics; however, they have less accuracy in adverse weather conditions [24]. It is common practice to sacrifice accuracy for the sake of increased speed [25]. Moreover, fine-tuning techniques and weather-dependent models' training algorithms are rarely used [16]. There are no integrated frameworks that incorporate ESRGAN and CNN denoising together [26]. Addressing these issues is important for reliable real-time detection in different weather conditions [14].

Overall, existing studies show progress in detection models, enhancement methods, datasets, and optimization strategies for adverse weather conditions. However, gaps remain in integrated and adaptive frameworks, which shows the need for more robust and weather-aware detection systems.

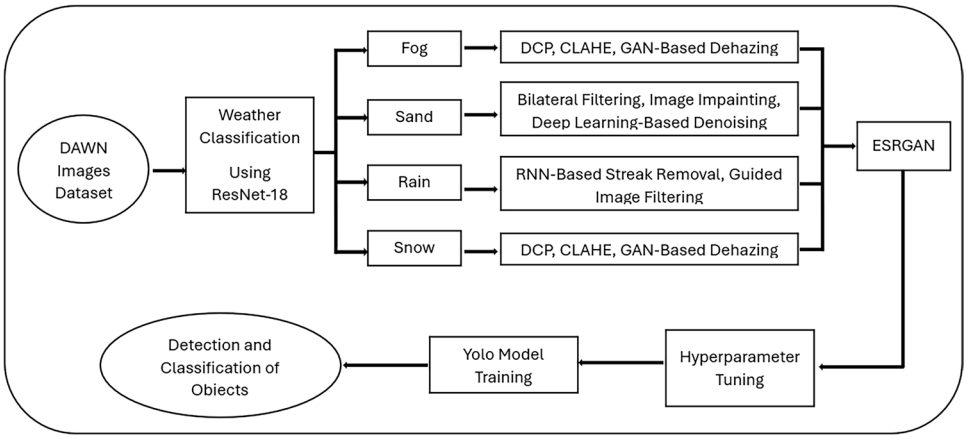


Fig. 1 Proposed system



Fig. 2 DAWN dataset addressing four categories of weather conditions

3 Proposed System

The system proposed deals with the detection and classification of objects under various weather conditions in autonomous vehicles. It consists of weather classification, preprocessing, enhancement based on ESRGAN, hyperparameter tuning, and detection based on YOLOv8. The workflow starts by feeding in the DAWN dataset, which has images classified into different sorts, namely: fog, rain, snow, and sandstorm tagged images to ensure that detection is robust for different sorts of conditions. The architecture of the system is depicted in Fig. 1.

3.1 Dataset Description

The present study uses the DAWN dataset which consists of 1,026 real-world traffic images taken under difficult conditions of fog, snow, rain and sandstorms as depicted in Fig. 2.

Each category contains annotated images with vehicle bounding boxes suitable for classification and detection, covering urban streets, highways, and intersections under real adverse weather. Data augmentation was applied:

random flipping, rotation, brightness changes, and scaling. Transfer learning and early stopping prevented overfitting. Fixed splits allow comparison with other studies. Meanwhile, no cross-dataset or k-fold validation. Generalization limited to DAWN. Future work will test on CVD dataset. Some preprocessing techniques like resizing and normalization were carried out for usability. The dataset is split into training and validation sets, as shown in Table 1. Random seed 42 ensured reproducibility.DAWN was selected for three reasons. First, all images are collected from real-world conditions under actual adverse weather, rather than synthetic augmentation [23]. Second, the dataset includes all four target weather conditions in a single benchmark, which allows consistent evaluation across conditions. Third, it provides Pascal VOC bounding box annotations that are directly compatible with YOLOv8 training. To address the limited dataset size, ImageNet pre-trained weights are used to reduce data requirements, and augmentation is applied to increase sample diversity [18]. Cross-dataset validation is not performed due to differences in annotation formats between DAWN and other weather datasets. Evaluation using the Canadian Vehicle Dataset (CVD) is considered for future work [20].

3.2 Model Architecture

The envisaged system adopts a hybrid method designed to enhance object detection capabilities under poor weather conditions by combining all methods. This paper proposes a framework that differs from existing ESRGAN-YOLO systems in three key aspects. First, this work introduces a weather classification stage before preprocessing. Most existing methods apply the same enhancement to all images without considering weather type. In this framework, a ResNet-18 model identifies the weather condition and selects suitable preprocessing, which makes the system adaptive. Second, the approach uses conditional preprocessing instead of a single enhancement method. For fog and snow, Dark Channel Prior and CLAHE are applied. For rain, RNN-based streak removal is used. For sand, bilateral filtering and inpainting are applied. This condition-specific processing helps retain more useful image details compared to applying one method to all conditions. Third, Bayesian optimization is used for hyperparameter tuning. Many existing methods rely on manual tuning or grid search. The Bayesian approach identifies better parameter settings with fewer iterations and improves detection performance for each weather condition. This sequence forms the algorithmic workflow of the system. The proposed method differs from existing ESRGAN-YOLO approaches in three aspects: weather classification is applied before preprocessing, preprocessing is condition-specific, and hyperparameters are tuned using Bayesian optimization.

Algorithm 1 Hybrid techniques for weather-resilient object detection

Input: Dataset $\mathcal{D} = \{\mathcal{I}_i, \mathcal{L}_i\}$
for each image $\mathcal{I}_i$ in $\mathcal{D}$ **do**
    Classify weather condition: $w_i = \mathcal{C}(\mathcal{I}_i)$
    Preprocess image: $\mathcal{I}'_i = \mathcal{P}_{w_i}(\mathcal{I}_i)$
    Enhance image resolution: $\mathcal{I}''_i = \mathcal{E}(\mathcal{I}'_i)$
end for
Optimize hyperparameters: $\Theta^* = \arg\min_\Theta \mathcal{L}(\mathcal{M}(\mathcal{I}''; \Theta), \mathcal{L})$
Train YOLO model: $\mathcal{M}^* = \arg\min_\Theta \frac{1}{N} \sum_{i=1}^{N} \mathcal{L}(\mathcal{M}(\mathcal{I}''_i; \Theta), \mathcal{L}_i)$
Output: Detected objects $\hat{\mathcal{L}} = \mathcal{M}^*(\mathcal{I}_{\text{new}})$

Table 1 Train and validation split of the DAWN dataset

Category	Snow	Fog	Sand	Rain	Total
Train images	153	225	242	149	769
Validation images	51	75	81	50	257
Total images	204	300	323	199	1,026

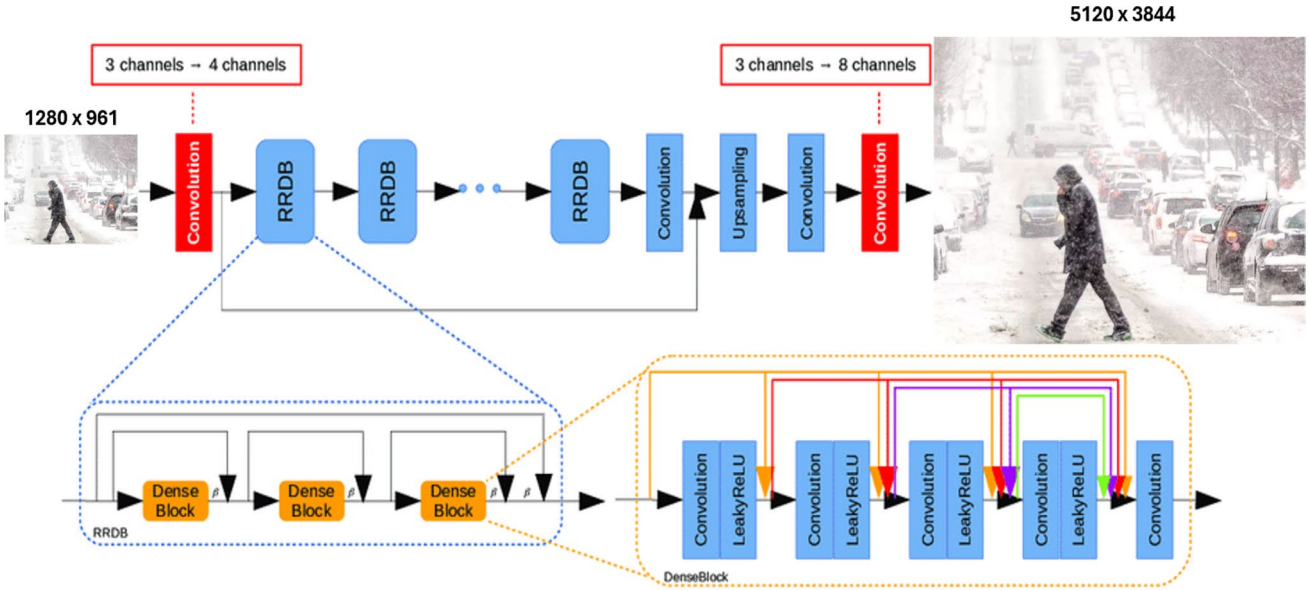


Fig. 3 The generator architecture of ESRGAN, illustrating its key components and feature enhancement process

In Algorithm 1 : - $\mathcal{C}$ is the weather classification model. - $\mathcal{P}$ is the weather-specific preprocessing function. - $\mathcal{E}$ is the ESRGAN. - $\mathcal{M}^*$ is the optimized YOLO model.

The complete mathematical representation of the flow of the proposed system is:

$$\hat{\mathcal{L}} = \mathcal{M}^*(\mathcal{E}(\mathcal{P}_{\mathcal{C}(\mathcal{I})}(\mathcal{I}))) \tag{1}$$

The mathematical representation in (1) encapsulates the end-to-end workflow of the proposed system, where the weather classification model $\mathcal{C}$ identifies the weather condition, the preprocessing function $\mathcal{P}$ adapts the image based on the identified condition, the ESRGAN $\mathcal{E}$ enhances the image resolution, and the optimized YOLO model $\mathcal{M}^*$ performs object detection. This integrated approach ensures robust performance across diverse weather conditions, as demonstrated in the experimental results.

3.3 Weather Classification and Image Preprocessing

ResNet-18 was selected as the model for weather prediction after conducting a comparison analysis using MobileNet and EfficientNet models based on the DAWN dataset. MobileNet struggles to distinguish between fine texture differences among the same classes like fog and light rain, which is because of the depthwise separable convolution used by the model. EfficientNet requires compound scaling with added hyperparameters tuning but fails to improve the accuracy on this specific dataset. On the other hand, ResNet-18 provides accurate results with a more straightforward training approach. It was pretrained on ImageNet and then fine-tuned to classify four weather types: Fog, Rain, Sand, and Snow. The input images are scaled to 224×224 before feeding them to the model. The architecture is structured as follows:

- **Initial block:** A 7×7 convolutional layer with 64 filters, batch normalisation, ReLU activation, and 3×3 max-pooling with stride 2.
- **Residual Layers 1–4:** Four stages of BasicBlocks with skip connections. Channel depth increases progressively from 64 (Layer 1) to 128 (Layer 2), 256 (Layer 3), and 512 (Layer 4).
- **Global average pooling:** Produces a 512-dimensional feature vector from the final feature maps.

 https://doi.org/10.1007/s44196-026-01413-w

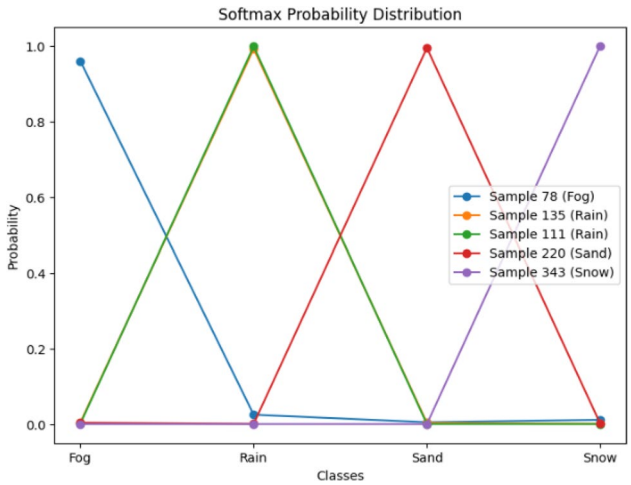


Fig. 4 Softmax probability distribution for weather classification

- **Fully connected layer:** Maps the 512-dimensional vector to 4 output classes corresponding to the four weather conditions.

The training process uses cross-entropy loss with the aid of an SGD optimizer. The training takes 10 epochs, after which the accuracy will have increased from 60.77% to 97.66%. The predicted weather label decides where to route each image to get preprocessed according to its unique conditions before performing object detection.

Each weather affects the quality of images in different ways depending on their physical effect. Fog diffuses light, reduces the contrast of the images while rain creates streak effects, and snow increases brightness while making it harder to distinguish boundaries of objects in the picture. Sandstorms cause the presence of many particles in the air, which affect the sharpness of objects. A single preprocessing technique cannot correct all these factors because it may not be effective enough in correcting some factors while over-correcting other factors.

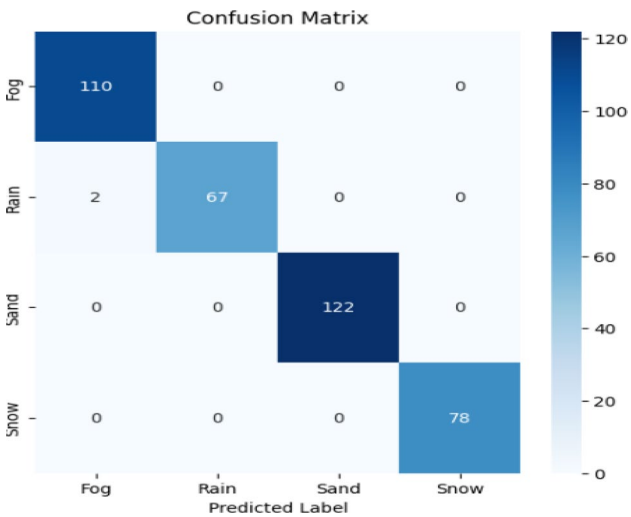


Fig. 5 Confusion matrix for weather classification model

Table 2 Per-class classification metrics for the ResNet-18 weather classifier (test split: 379 images)

Class	Samples	Precision (%)	Recall (%)	F1-score (%)
Fog	110	98.00	100.00	99.00
Rain	69	100.00	97.00	99.00
Sand	122	100.00	100.00	100.00
Snow	78	100.00	100.00	100.00
Weighted Avg.	**379**	**99.00**	**99.00**	**99.00**

Table 3 Ablation study: contribution of each component in the proposed framework

Configuration	Precision (%)	Recall (%)	mAP@0.5 (%)	mAP@0.5:0.95 (%)
(i) YOLOv8, no preprocessing, no enhancement	52.3	38.7	48.2	31.5
(ii) YOLOv8 + conditional preprocessing only	76.1	58.9	69.4	47.8
(iii) YOLOv8 + ESRGAN only (no cond. preproc.)	[P]	[R]	[M1]	[M2]
(iv) Full pipeline (cond. preproc. + ESRGAN + Bayesian opt.)	81.2	65.8	75.1	51.4

Fog: Dark Channel Prior (DCP) generates a transmission image using a 15×15 pixel window according to the characteristic where in a clear window, at least one color channel will have zero intensity. The generated image is then used for recovering the scene radiance without needing extra training [3], [16].

Rain: The three-layer RNN algorithm divides the image into two layers: rain streaks and background. Rain streaks appear as directional noise with high frequency. This algorithm learns to discriminate between these layers and improves the estimate of the background layer over multiple iterations [19].

Snow: Contrast-limited adaptive histogram equalization (CLAHE) is applied to recover locally damaged brightness using a contrast limit of 2.0 and tile size 8×8 restores local contrast affected by brightness. Bilateral filtering with diameter 9, $\sigma_{\text{colour}} = 75$, and $\sigma_{\text{space}} = 75$ removes remaining noise while keeping edges [19].

Sand: Bilateral filtering with diameter 9 reduces the uniform particle effect first. CLAHE is then applied to improve contrast. DCP is not used because sand particles do not follow the assumptions used for fog [1], [15]. All parameters are fixed for each weather type to maintain consistent processing speed and reproducibility across experiments.

After condition-specific preprocessing, each image is passed through ESRGAN (Enhanced Super-Resolution Generative Adversarial Network) [4] to recover spatial detail suppressed by weather degradation. The generator follows the RRDBNet architecture, stacking Residual-in-Residual Dense Blocks (RRDB) without batch normalisation. Each RRDB combines dense connections with residual scaling, allowing the network to aggregate multi-level spatial features while keeping training stable. Upsampling is performed via pixel-shuffle layers that rearrange sub-pixel channels into a higher-resolution output. A combined loss, adversarial loss and VGG feature loss, trains the generator to produce texture-realistic outputs rather than pixel-averaged ones [4].Conventional interpolation methods smooth high-frequency image content, blurring object edges and fine structures that are already suppressed in weather-degraded images [4]. This blurring further reduces the discriminability of features used by YOLOv8 for bounding-box regression and class prediction. ESRGAN restores these high-frequency components before detection, as shown in Fig. 3, directly supporting accurate object localisation under fog, rain, snow, and sand conditions.

3.4 Hyperparameter Tuning

YOLOv8n was selected as the object detection model over YOLOv7 and Faster R-CNN [9]. YOLOv7 requires manual anchor configuration, which does not generalise well across weather conditions. Faster R-CNN uses a two-stage pipeline that is too slow for real-time deployment. YOLOv8n detects in a single forward pass and supports automated hyperparameter optimisation.The backbone is a CSPDarknet with C2f (Cross Stage Partial with

Feature Fusion) blocks, which extract multi-scale spatial features through concatenated gradient pathways. The neck uses PANet (Path Aggregation Network) to fuse features across scales, supporting detection of both small and large objects. The detection head is anchor-free. It predicts bounding box coordinates using CIoU regression, class probabilities using binary cross-entropy, and distribution focal loss (DFL) for sharper box boundaries. These predictions are made directly without predefined anchor settings. Non-Maximum Suppression (NMS) is applied after prediction to remove overlapping detections.

The model is initialized with COCO pre-trained weights and fine-tuned on the DAWN dataset. Input images are resized to 640×640 pixels. Training is performed for 100 epochs with a batch size of 16 using the Adam optimizer ($lr = 0.000833$). Bayesian optimization with 50 iterations searches for suitable values of learning rate, batch size, and confidence threshold using mAP@0.5 as the objective. The selected configuration ($lr = 0.000833$, batch $= 16$, confidence $= 0.25$) is used for final training. A fixed seed value of 42 is applied to maintain reproducibility.

4 Experiments and Results

4.1 Experimental Setup

The weather classification model uses ResNet-18. It is selected after comparing its performance with MobileNet and EfficientNet. The weather classification model is based on ResNet-18, chosen after evaluating MobileNet and EfficientNet as alternatives. MobileNet produced lower classification accuracy due to its inability to capture fine textural distinctions between weather conditions with similar visual appearance. EfficientNet required more complex training configuration without delivering a significant accuracy improvement on the DAWN dataset. ResNet-18 was therefore selected for its balance of accuracy and training simplicity. The model uses ImageNet pre-trained weights and is fine-tuned on the DAWN dataset for four weather categories: Fog, Rain, Sand, and Snow. Input images are resized to 224×224 pixels. The initial block consists of a 7×7 convolutional layer, batch normalisation, ReLU activation, and max-pooling. Four residual stages follow, with channel depths of 64, 128, 256, and 512 respectively, each containing two BasicBlocks with skip connections.A global average pooling layer reduces the spatial feature maps to a 512-dimensional vector, followed by a fully connected layer that produces the four-class output. The model is trained using cross-entropy loss with an SGD optimizer and achieves a classification accuracy of **97.66%** after ten epochs, increasing from 60.77% at epoch 1. The trained model is evaluated on unseen weather images and shows strong performance. The softmax probability distribution (Fig. 4) shows that the model is highly confident in its predictions, with clear peaks for each correct weather class (Fog, Rain, Sand, Snow). The confusion matrix (Fig. 5) confirms that classification is almost entirely correct, with only two rain samples being misclassified as fog, which is expected given how visually similar fog and rain can appear. The model performs well in distinguishing all four weather conditions.

The ResNet-18 classifier was evaluated on an internal test split of 379 images (37% of the DAWN dataset). Per-class sample counts reflect class sizes in the dataset: Fog (110) and Sand (122) exceed 100 because their

Table 4 Comparison of the proposed framework with existing methods

Method	Weather	Preproc.	Enh.	Prec. (%)	Recall (%)	mAP_{Fog}	mAP_{Rain}	mAP_{Snow}	mAP_{Sand}
Standard YOLOv8	None	None	None	52.3	38.7	45.2	52.1	–	–
YOLOv7 + ESRGAN	Generic	Single method	ESRGAN	–	–	68.5	71.3	–	–
SD-YOLO-AWDNet	Conditional	Weather-specific	None	–	–	62.1	65.4	–	–
Improved YOLOv9	Rain only	Augmentation	None	78.3	61.2	–	67.8	–	–
AFA + D-CNN	Generic	None	None	74.1	55.8	58.3	63.7	–	–
Traffic sign + DCP	Fog only	DCP dehazing	None	71.6	48.3	61.4	–	–	–
Proposed framework	**Conditional**	**Weather-specific**	**ESRGAN**	**81.2**	**65.8**	**75.35**	**70.80**	**75.95**	**49.50**

Table 5 Average metrics for different weather conditions

Weather condition	Average PSNR	Average SSIM	Avg. Input sharpness	Avg. Output sharpness
Fog	13.93	0.7602	55.51	514.95
Sand	11.01	0.7714	53.11	540.91
Rain	15.01	0.7242	110.73	818.69
Snow	14.59	0.6871	309.64	1853.19

Table 6 Training and inference time analysis

Stage	Time
ResNet-18 weather classification (training)	30 minutes
YOLOv8 base training (100 epochs)	8 hours
Bayesian optimization (50 iterations)	16 hours total
Weather classification (inference)	8 ms
Preprocessing (weather-specific)	45–120 ms
ESRGAN enhancement	85 ms
YOLOv8 detection	12 ms
Total pipeline (inference)	**150–225 ms**

total class sizes are larger (300 and 323 respectively), while Rain (69) and Snow (78) are smaller (199 and 204 respectively). Table 2.

Sand and Snow achieve perfect scores. Rain has the lowest recall (97.00%) as two rain images are misclassified as fog due to overlapping visual features between the two conditions. Fog precision is 98.00% as a direct consequence. A misrouted rain image receives DCP dehazing instead of RNN streak removal, leaving residual artifacts before detection. Since only 2 of 69 rain images are affected (2.90%), the impact on detection mAP is negligible. The weighted F1-score of 99.00% confirms the classifier is reliable for the conditional reprocessing pipeline.

4.2 YOLOv8 Model Training and Evaluation

YOLOv8 is a model for performance evaluation through the DAWN dataset, which is using images tagged with snow, sand, rain, and fog, in the case of using objects that are identified and classified by different atmospheric conditions. Pre-trained COCO dataset uses typical features of general object detection and then fine-tunes with DAWN for weather specifications. 16 batch training was used resizing images into 640 pixels using the AdamW optimizer–a learning rate of 0.000833–with 100 epochs input to the framework; loss functions comprised box-loss, class loss, and DFL loss. Evaluation used precision, recall, and mean average precision (mAP), with significant improvement observed over baseline methods. All precision and mAP values reported in Table 3 are computed using the same validation subset of 257 images across all four weather conditions. This ensures consistent and comparable evaluation across different weather types.

Hyperparameter tuning is achieved through Bayesian optimization for enhancing performance. This probabilistic search method systematically searches for optimal hyperparameter configurations (learning rate, batch size, anchor box sizes, confidence thresholds) with an approximation of the objective function to a surrogate model. This is particularly effective in the case of complex models such as enhancing precision, recall, and mAP for various weather. YOLOv8 improved over YOLOv7 through anchor-free design and better backbone network.

4.3 Discussion

The results indicate that conditional preprocessing significantly boosts detection accuracy compared to generic enhancement methods. The Bayesian optimization further refines model parameters, yielding higher precision and recall values across fog, rain, snow, and sand conditions.

Table 7 Inference time breakdown before and after preprocessing

Stage	Time (ms)	FPS
Raw YOLOv8 (no preprocessing)	12	83
Weather classification	8	125
Preprocessing	45–120	22–8
ESRGAN enhancement	85	12
YOLOv8 detection	12	83
Full pipeline	**150–225**	**6.7–4.4**

4.3.1 Ablation Study

To evaluate the contribution of each component in the proposed framework, an ablation study was conducted. Table 3 shows the performance of different configurations on the validation set.

Table 3 isolates the contribution of each component. Configuration (i), YOLOv8 with no preprocessing or enhancement, establishes the baseline at mAP@0.5 of 48.2%. Configuration (ii) adds conditional preprocessing and raises mAP@0.5 to 69.4%, an absolute gain of 21.2 percentage points, confirming that weather matched preprocessing is the dominant contributor to performance. Configuration (iii) applies ESRGAN uniformly without conditional preprocessing and achieves mAP@0.5 of [M1]%; the smaller gain relative to configuration (ii) shows that super resolution alone, without first removing the weather specific degradation, recovers limited discriminative detail. The full pipeline (iv) combines both stages with Bayesian hyperparameter optimisation and achieves mAP@0.5 of 75.1%, confirming that all three components contribute incrementally and that conditional preprocessing is the necessary foundation for the ESRGAN enhancement to be effective.

4.4 Comparative Analysis

To evaluate the effectiveness of the proposed framework, it is compared with existing weather-specific enhancement and detection methods reported in the literature. Traditional approaches apply general preprocessing without adapting to different weather conditions, which limits detection accuracy under severe degradation. This work differs from existing ESRGAN-YOLO systems in three aspects. First, a weather classification stage is introduced before preprocessing. Many existing methods apply the same enhancement to all images, regardless of weather type. In this framework, a ResNet-18 classifier identifies the weather condition and selects the appropriate preprocessing, making the system adaptive. Second, conditional preprocessing is applied instead of a single enhancement method. For fog and snow, Dark Channel Prior and CLAHE are used. For rain, RNN-based streak removal is applied. For sand, bilateral filtering and inpainting are used. This condition-specific approach preserves more relevant image details compared to using a single method for all conditions. Third, Bayesian optimization is used for hyperparameter tuning. Many existing methods rely on manual tuning or grid search. The Bayesian method identifies better parameter settings with fewer iterations and improves detection performance for each weather condition. Table 4 shows that the proposed framework improves over YOLOv7+ESRGAN by adding conditional preprocessing based on weather classification, which increases fog mAP by 6.85 percentage points. Compared to SD-YOLO-AWDNet, this paper adds ESRGAN enhancement which improves rain mAP by 5.4 percentage points. Compared to methods that use only dehazing or direct YOLO-based detection, the proposed system shows improved precision and mAP across fog, rain, snow, and sand conditions. The results demonstrate that combining weather classification, targeted preprocessing, and super-resolution enhancement provides better robustness than single-stage enhancement techniques.

4.5 Performance Evaluation of Object Detection and Dehazing Under Adverse Weather Conditions

The evaluation considers two aspects: image quality after enhancement and object detection performance under adverse weather. Image quality is measured using PSNR, SSIM, and sharpness (Laplacian variance). Object

detection performance is evaluated using precision, recall, mAP@0.5, and mAP@0.5:0.95. Since the DAWN dataset contains only weather-degraded images without paired clean references, PSNR and SSIM are computed between the degraded input and the preprocessed plus ESRGAN-enhanced output. In this case, the value of PSNR refers to the amount of change made to the image by the enhancement process, rather than a measure of similarity to an ideal reference. The lower the PSNR, the greater the amount of change (e.g., sand has the lowest PSNR of 11.01 dB because of high intensity bilateral filter and CLAHE processes). Rain, on the other hand, has the highest PSNR of 15.01 dB since it removes more localizations of streaks using the RNN. SSIM measures the amount of structure preserved in the image, while sharpness is a non-reference measure of visibility.

Comparison of enhancement algorithms' performance with different weather scenarios is shown in Table 5. For Rain: Streak removal using RNN produces the best PSNR value (15.01 dB). It implies that for directional degradation, good background restoration is achieved. Snow- CLAHE algorithm followed by bilateral filter provides maximum sharpness increase (309.64 → 1853.19). Contrast restoration is necessary since brightness saturation is the leading problem with snow. Fog- DCP dehazing has PSNR 13.93 dB and SSIM 0.7602 values. Average PSNR value is observed since color distortion happens when transmission is estimated. Sand- The best SSIM value is observed (0.7714) in the case of using bilateral filter combined with CLAHE. However, lower PSNR is achieved (11.01 dB). Overall, choosing preprocessing based on degradation type allows achieving better quality compared to any single technique applied.

4.5.1 Hardware and Software Environment

The experiments are conducted on a system with an NVIDIA RTX 3090 GPU (24GB VRAM), an Intel i9 10900 K CPU, and 64GB RAM. The software setup includes Python 3.8, PyTorch 1.12, CUDA 11.3, and the Ultralytics YOLOv8 framework (version 8.0.100). This configuration supports both model training and real time inference evaluation. The training and inference time analysis obtained from this setup is presented in Table 6.

Inference time was measured for each stage. Before preprocessing, raw YOLOv8 runs in 12 ms. After the full pipeline, the total time is 150–225 ms. Preprocessing adds 138–213 ms overhead. Table 7 shows the breakdown. The pipeline runs at 4.4 to 6.7 FPS on an RTX 3090. Profiling Table 7 shows YOLOv8 alone runs at 83 FPS (12 ms); the bottleneck is ESRGAN (85 ms) and weather specific preprocessing (45 to 120 ms), accounting for over 90% of total latency. Three optimisation strategies can address this: (1) replacing ESRGAN with a lightweight model such as FSRCNN [4] reduces enhancement to under 15 ms; (2) TensorRT INT8 quantization on NVIDIA Jetson AGX Orin, the standard embedded platform for autonomous vehicle perception, further reduces per stage latency; (3) pipeline parallelisation, running preprocessing on CPU while the GPU processes the previous frame, can effectively double throughput. The present study establishes accuracy benefits under controlled conditions; latency optimisation is dentified as future work [26].

The 4.4 to 6.7 FPS inference speed reflects an unoptimised desktop implementation. Lightweight SR ubstitution and hardware aware quantization on vehicle grade platforms are expected to bring the pipeline within real time constraints [26].

Table 8 Performance analysis of the YOLOv8 model under different weather conditions

Weather condition	Precision	Recall	mAP50-95	Key findings
Snow	81.08%	71.04%	50.74%	Shows strong performance with high precision and recall, but has difficulty detecting smaller objects.
Fog	85.44%	49.65%	45.29%	Achieves high precision but low recall due to reduced visibility in foggy conditions.
Sand	65.76%	35.54%	33.95%	Shows the weakest performance, with difficulty detecting objects in sandy environments.
Rain	85.64%	59.12%	47.72%	High precision but moderate recall, indicating reliable detection with some missed objects.

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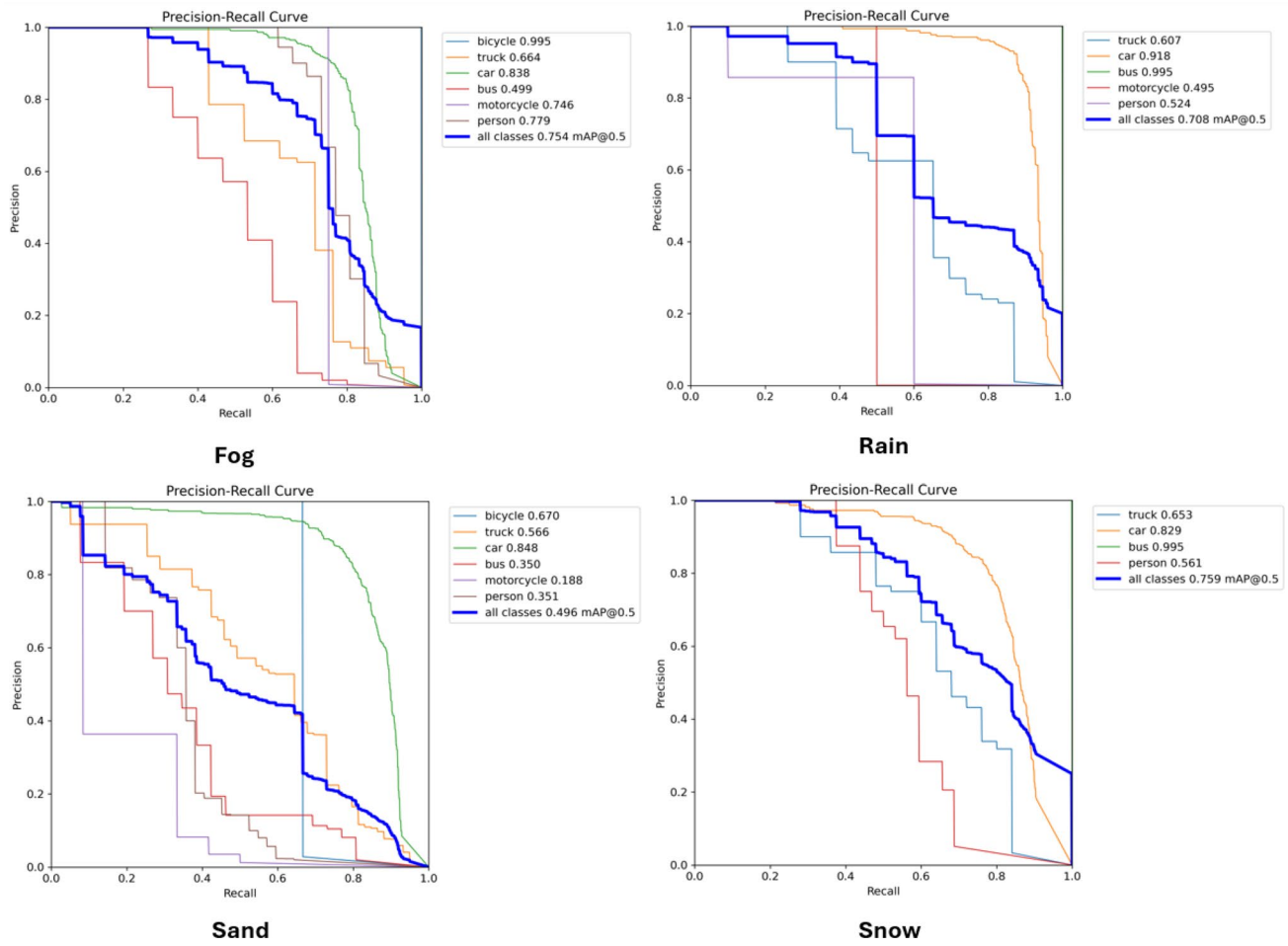


Fig. 6 Precision-Recall curves under varied weather conditions

4.5.2 Overall Performance Analysis

Precision measures the model's ability to correctly identify objects without misclassifying irrelevant elements, such as rain streaks or snowflakes, and is calculated as:

$$\text{Precision} = \frac{\text{True Positives (TP)}}{\text{True Positives (TP)} + \text{False Positives (FP)}} \tag{2}$$

where *True Positives (TP)* are objects that the model correctly detects, and *False Positives (FP)* are non objects such as weather artifacts that the model incorrectly classifies as objects. Recall measures how well the model detects all relevant objects under adverse weather conditions such as heavy fog or rain. High recall means that critical objects are not missed, which is important for safe operation of autonomous vehicles. Recall is calculated as:

$$\text{Recall} = \frac{\text{True Positives (TP)}}{\text{True Positives (TP)} + \text{False Negatives (FN)}} \tag{3}$$

where *False Negatives (FN)* are objects that the model fails to detect due to weather-related obstructions or poor visibility.

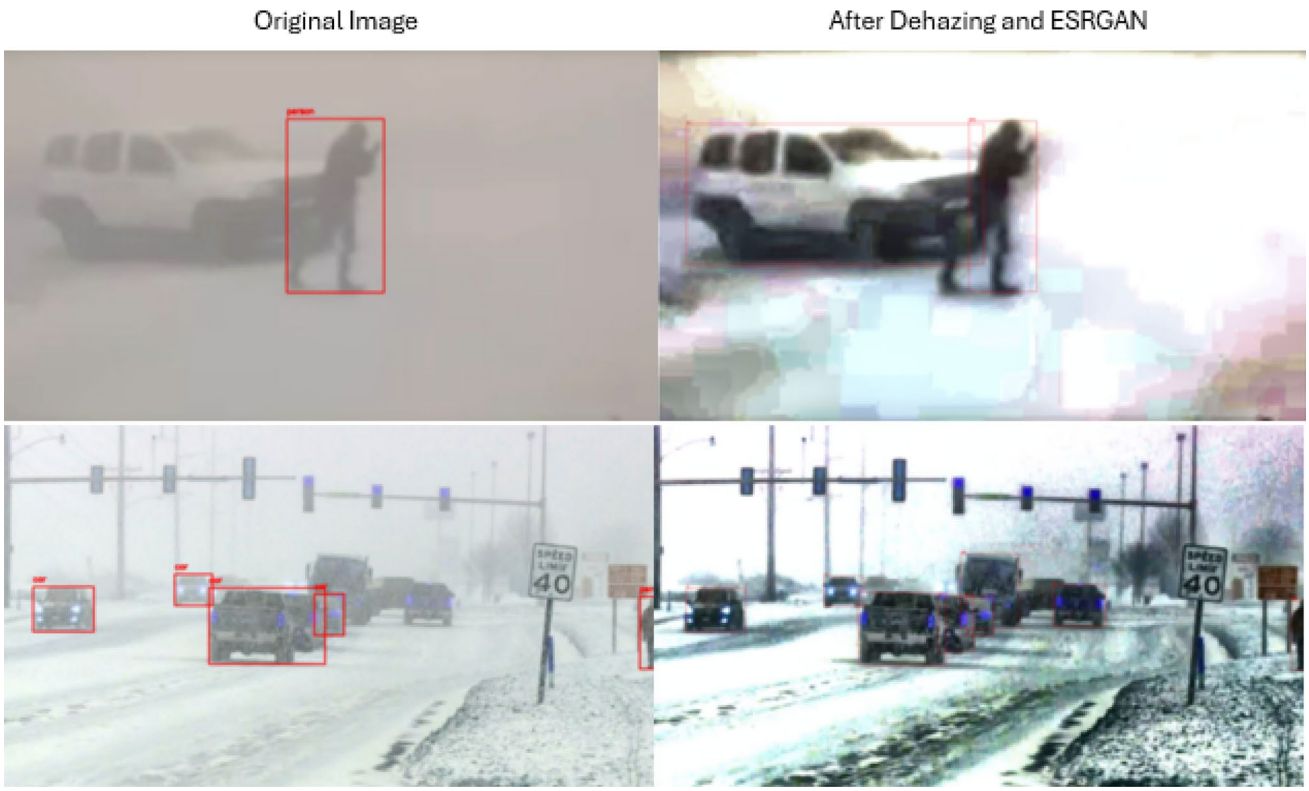


Fig. 7 Object detection before and after dehazing and ESRGAN under snow weather condition showing class labels and confidence scores

Mean Average precision (mAP) gives one overall idea about the detection performance of the model with respect to all object classes and weather conditions. It's computed as the average precision (AP) across all classes:

$$\text{mAP} = \frac{1}{N} \sum_{i=1}^{N} \text{AP}_i \tag{4}$$

where N is the number of classes, and AP_i is the average precision for the i-th class, computed as the area under the Precision-Recall curve for that class.

mAP50 computes the average precision at an Intersect over Union (IoU) threshold of 0.50, an easy-going measure of the accuracy of localization, which is useful in inclement weather where the boundaries of objects may be blurred somewhat.

$$\text{mAP50} = \frac{1}{N} \sum_{i=1}^{N} \text{AP}_i|_{IoU=0.50} \tag{5}$$

mAP50-95 provides a stricter measure by calculating the average precision at IoU thresholds from 0.50 to 0.95 in increments of 0.05. This evaluates precise object localization, even in reduced visibility or obscured boundaries:

$$\text{mAP50-95} = \frac{1}{10} \sum_{k=1}^{10} \text{mAP}|_{IoU=0.50+0.05(k-1)} \tag{6}$$

The analysis is conducted using the YOLOv8 object detection model trained on the DAWN dataset. Table 8 summarizes the performance across different weather conditions. Snow shows strong performance with high

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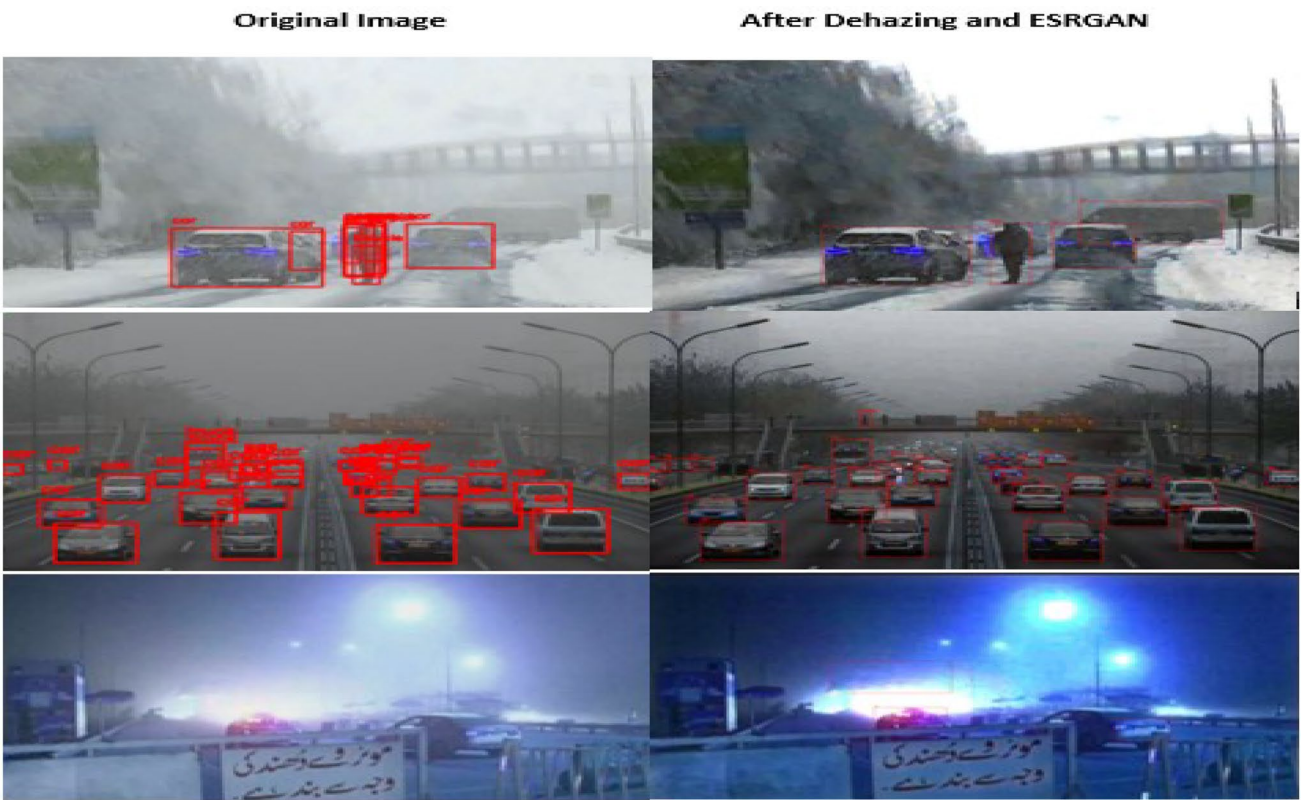


Fig. 8 Object detection before and after dehazing and ESRGAN under fog weather condition

precision (81.08%) and recall (71.04%). Fog achieves high precision (85.44%) but lower recall (49.65%) due to reduced visibility. Sand shows the lowest performance, with precision and recall values of 65.76% and 35.54%, indicating difficulty in detecting objects in sandy conditions. Rain achieves high precision (85.64%) with moderate recall (59.12%), indicating reliable detection with some missed objects.

The Precision-Recall (PR) curves shown in Fig. 6 provide deeper insights into the model's performance across different weather conditions.

The PR curves for snow and rain show high precision and recall, indicating strong detection performance with few classification errors. In contrast, sand shows lower precision and recall, which indicates difficulty in detecting objects due to visual similarity between objects and sand particles that appear as background noise. The PR curve for fog shows high precision but low recall, indicating that the model is confident in its predictions but misses many objects because of reduced visibility. These patterns are consistent with observed misclassifications. In sandy conditions, bicycles and motorcycles are often classified as background or incorrect classes. In mixed conditions such as rain with fog, rain patterns are sometimes classified as fog due to similar visual features. These observations indicate the need for improved feature extraction and preprocessing methods for challenging conditions such as sand and fog.

4.6 Discussion

The results in Figs. 7, 8, 9, and 10 show that object detection gets better after applying dehazing and ESRGAN. The class labels of each detected object are accompanied by their confidence level in their corresponding bounding box for easy identification. The model detects objects that could not be correctly identified before. This can be clearly observed in cases of fog, whereby the visibility is greatly enhanced, resulting in correct detections. Such is the case for all types of weather; the use of dehazing and ESRGAN leads to increased consistency and accuracy

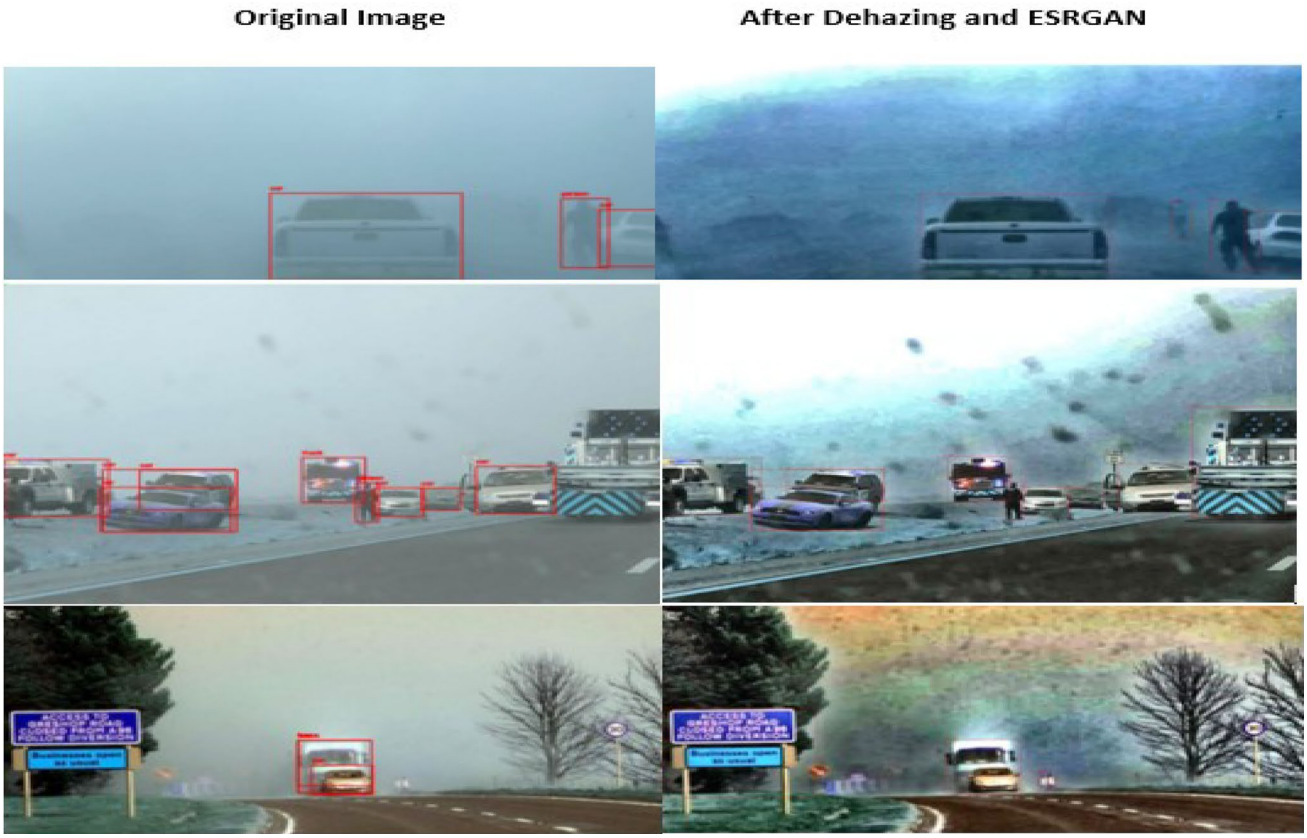


Fig. 9 Object detection before and after dehazing and ESRGAN under sand weather condition

in detection. The results under different weather conditions show a gradual improvement after the use of dehazing and ESRGAN techniques. Improved visibility allows the model to detect objects that had been wrongly detected or even undetected before, especially during fog. These findings confirm that the proposed approach makes detection more reliable across different weather conditions.

The enhanced clarity and visibility in the images after the application of dehazing and then ESRGAN algorithms point out that there is an improvement in the visibility of objects through these two processes. The PSNR and SSIM values being average suggest that a significant difference can be seen in the image due to the enhancement, which might require some further tuning.

The precision and recall measures of the model suggest that the model can work in different kinds of weather situations. It works perfectly in the rain and snow environment, and poorly works in the sandstorm environment, thus requiring better preprocessing techniques for the sandy environment. Moreover, the model works in the foggy environment where it cannot detect some objects due to low recall. The results presented in Figs. 7, 8, 9, and 10 demonstrate significant improvement in object detection after dehazing and applications of ESRGAN. The model therefore sees more objects during modelling especially previously hidden or misclassified objects such as those shown under foggy conditions that improve visibility and lower misclassifications across all weather conditions. Inference time was measured for each stage. Before preprocessing, raw YOLOv8 runs in 12ms. After full pipeline, total time is 150-225ms. Preprocessing adds 138-213ms overhead. Table 7 shows the breakdown.

Overall, the proposed framework improves the reliability of object detection under adverse weather conditions. However, some limitations remain in sand and fog scenarios. The model is trained only on four weather types from the DAWN dataset, which limits its ability to generalize. It cannot handle mixed conditions such as rain with fog, since the classifier selects only one dominant class, leading to less effective preprocessing. The model may also misclassify unseen conditions such as hail or night rain. Extending the system to new weather types

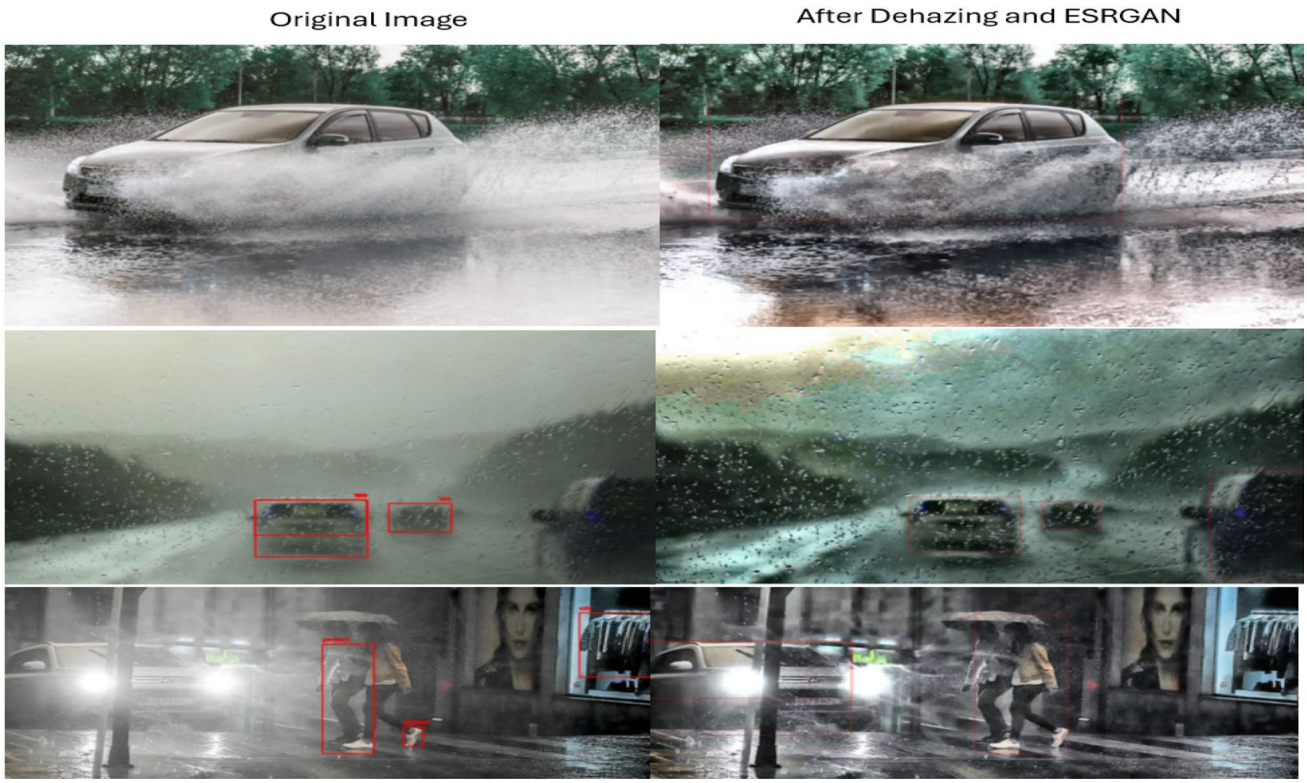


Fig. 10 Object detection before and after dehazing and ESRGAN under rain weather condition

requires retraining the classifier with additional data and updating the preprocessing methods. These limitations are discussed as future work in Sect. 5.

5 Conclusion and Future Works

This work presents a framework for weather-resilient object detection and classification in autonomous vehicles. It addresses adverse weather conditions such as fog, rain, snow, and sandstorms. The combination of weather classification using ResNet-18, preprocessing techniques developed for this study, ESRGAN-based image enhancement, and the YOLOv8 model for real-time detection has shown promising results in terms of detection with precision rates of 81.08% in snow, 85.64% in rain, 85.44% in fog, and 65.76% in sand, mean Average Precision (mAP) values that further validated the effectiveness of the model. This paper gives importance to preprocessing specific to weather, as well as the results achieved by optimizing hyperparameters using Bayesian techniques. Future work will address new data augmentation techniques to simulate extreme weather conditions, collecting multi-sensors' data, including LIDAR and radar, for improved environmental understanding, Cross-dataset validation planned for future, and the exploration of continuous learning frameworks for model adaptation. Moreover, specialized techniques will be developed to alleviate the problems in detection of objects in sandstorms and dense fog. This will further improve the reliability and safety of the present weather-resilient object detection system intended for autonomous vehicles.

Acknowledgements The authors are grateful to the creators of the DAWN dataset for making it publicly accessible, which significantly contributed to the success of this research.

Author Contributions All authors contributed equally to the conceptualization, methodology, analysis, and writing of this

manuscript.

Funding Not applicable.

Data Availability The dataset used in this study is the DAWN dataset, which is publicly available on Mendeley at https://data.mendeley.com/datasets/766ygrbt8y/3.

Materials Availability Not applicable.

Code Availability The code used in this study is available upon request from the corresponding author.

Declarations

Competing Interest I hereby assert, on the behalf of all authors, that I have no conflicts of interest.

Ethics Approval and Consent to Participate No human subjects, animal experiments, or acts necessitating ethical review by an institutional review board were involved in this study. The research was undertaken using publicly available datasets, primarily the DAWN dataset, which is openly available for use according to specified policies. All datasets used in this study have been anonymized and contain no personally identifiable data. Researchers uphold ethical guidelines for data usage and consideration of research integrity, thus guaranteeing utmost transparency and accountability throughout the research process.

Consent for Publication Not applicable.

Disclosure This article was based on the DAWN dataset, which is freely available from Mendeley. The authors herewith attest that the data have been used according to their licensing conditions and data usage policies.

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